Experiment 1: Case Study on the Analysis of SDLC Models Applied to a Real-World Software Application

Submitted by Shive Shangkarran S

Register Number: 241201050

III Year – Mechatronics Engineering

Title

Case Study on the Analysis of SDLC Models Applied to Uber

Aim

To study different Software Development Life Cycle (SDLC) models and determine the most suitable model for the development and maintenance of the **Uber** ride-hailing application.

Introduction

Software Development Life Cycle (SDLC) is a systematic process used to develop, test, deploy, and maintain software applications. Selecting the appropriate SDLC model ensures efficient project execution, high software quality, and customer satisfaction.

Uber is one of the world's leading ride-hailing platforms, connecting passengers with drivers through a mobile application. The platform continuously evolves by introducing new features, improving security, enhancing user experience, and expanding services. Therefore, selecting an appropriate SDLC model is essential for its successful development and maintenance.

Application Overview

Uber is a globally recognized ride-hailing and transportation platform that connects passengers with nearby drivers through a mobile application. The platform enables users to book rides conveniently, track their driver in real time using GPS, and make secure online payments. In addition to ride booking, Uber offers features such as fare estimation, ride scheduling, trip history, driver registration, ratings and reviews, push notifications, and emergency safety features. The application is designed to provide a seamless, safe, and efficient transportation experience while ensuring high availability and reliability for millions of users worldwide.

Major Features

- User Registration and Login
- Ride Booking

- GPS-Based Driver Tracking
- Fare Estimation
- Online Payment
- Ride Scheduling
- Driver Registration
- Ratings and Reviews
- Trip History
- Push Notifications
- Emergency Safety Features

Business Requirements

To provide a reliable and user-friendly transportation service, Uber must satisfy several key business requirements. The application should enable passengers to book rides instantly and accurately match them with nearby available drivers. It must support secure and seamless online payment methods while providing precise GPS-based navigation and real-time ride tracking. Since Uber serves millions of users across different locations, the platform should be highly scalable, reliable, and capable of handling heavy traffic without compromising performance. Furthermore, the application should continuously evolve by incorporating user feedback, improving existing features, and introducing innovative services to enhance customer satisfaction.

Key Business Requirements

- Provide real-time ride booking.
- Match passengers with nearby drivers.
- Ensure secure online payments.
- Deliver accurate GPS navigation.
- Handle millions of users simultaneously.
- Continuously improve based on user feedback.

Analysis of SDLC Models

1. Waterfall Model

Description

The Waterfall Model follows a sequential development process where one phase is completed before moving to the next.

Advantages

- Simple and structured
- Easy documentation
- Well-defined milestones

Disadvantages

- Difficult to accommodate changing requirements
- Late testing phase
- Less suitable for dynamic applications

Suitability for Uber

Since Uber frequently updates its application with new features and improvements, the Waterfall Model is **not suitable**.

2. Incremental Model

Description

The software is developed in small functional modules, with each module delivered separately.

Advantages

- Faster release of features
- Easier debugging
- Lower development risk
- Continuous customer feedback

Disadvantages

- Requires careful integration
- Planning becomes more complex

Suitability for Uber

Features such as ride booking, online payments, and ride scheduling can be developed incrementally, making this model a good choice.

3. Spiral Model

Description

The Spiral Model combines iterative development with continuous risk analysis.

Advantages

- Excellent risk management
- Suitable for large projects
- Continuous customer evaluation
- Handles changing requirements effectively

Disadvantages

- High development cost
- Complex management

Suitability for Uber

Since Uber deals with sensitive user information, secure payments, and large-scale operations, the Spiral Model effectively manages technical and business risks.

4. Agile Model

Description

The Agile Model divides development into short iterations called sprints, allowing continuous delivery and frequent customer feedback.

Advantages

- Highly flexible
- Quick implementation of new features
- Faster bug fixing
- Continuous testing
- Improved collaboration

Disadvantages

- Requires experienced teams
- Continuous customer involvement

Suitability for Uber

Uber regularly introduces new services, improves navigation algorithms, enhances safety features, and fixes bugs. Agile supports these continuous improvements efficiently.

Comparison of SDLC Models

SDLC Model	Flexibility	Risk Management	Development Speed	Suitability for Uber
Waterfall	Low	Low	Slow	Not Suitable
Incremental	High	Medium	Fast	Suitable
Spiral	High	Very High	Medium	Highly Suitable
Agile	Very High	High	Very Fast	Most Suitable

Selected SDLC Model: Agile

Reasons for Selecting Agile

- Supports frequent application updates.

- Enables continuous customer feedback.
- Allows rapid feature development.
- Facilitates quick bug fixes and performance improvements.
- Improves collaboration among developers, testers, designers, and stakeholders.
- Supports Continuous Integration and Continuous Deployment (CI/CD).

SDLC Phases Applied to Uber

SDLC Phase	**Activities**
Requirement Analysis	Gather requirements from passengers, drivers, and business stakeholders.
Planning	Define project scope, schedule, resources, and budget.
System Design	Design application architecture, database, APIs, and user interfaces.
Development	Implement ride booking, GPS integration, payment gateway, and driver management modules.
Testing	Perform unit, integration, system, performance, security, and usability testing.
Deployment	Release updates through mobile app stores and cloud infrastructure.
Maintenance	Monitor performance, fix bugs, improve security, and introduce new features.

Challenges Faced During Development

- Real-time GPS tracking accuracy.
- Matching passengers with nearby drivers efficiently.

- Secure online payment processing.
- Handling peak-hour traffic and high user loads.
- Ensuring user and driver safety.
- Maintaining application availability across different regions.

Benefits of Agile for Uber

- Faster release of new features.
- Improved customer satisfaction through regular updates.
- Reduced development risks.
- Better collaboration among development teams.
- Continuous software testing and quality improvement.
- Easy adaptation to changing market demands.

Outcome

The analysis shows that while the Incremental and Spiral Models provide valuable benefits, the **Agile SDLC Model** is the most suitable for Uber. Its flexibility, iterative development, and continuous customer feedback enable Uber to deliver reliable, scalable, and user-focused transportation services.

Conclusion

Uber operates in a dynamic and highly competitive environment where customer expectations, technology, and business requirements evolve rapidly. The Agile SDLC Model provides the flexibility needed to adapt to these changes through continuous development, testing, and deployment. By adopting Agile practices, Uber can maintain high software quality, quickly introduce innovative features, and enhance the overall user experience while ensuring system reliability and security.